economic and political models we have inherited from the past are inadequate for dealing with such a challenge.

Take, for example, communism. As automation threatens to shake the capitalist system to its foundation, one might suppose that communism could make a comeback. But communism was not built to exploit that kind of crisis. Twentieth-century communism assumed that the working class was vital for the economy, and communist thinkers tried to teach the proletariat how to translate its immense economic power into political clout. The communist political plan called for a working-class revolution. How relevant will these teachings be if the masses lose their economic value, and therefore need to struggle against irrelevance rather than against exploitation? How do you start a working-class revolution without a working class?

Some may argue that humans could never become economically irrelevant, because even if they cannot compete with AI in the workplace, they will always be needed as consumers. However, it is far from certain that the future economy will need us even as consumers. Machines and computers could do that too. Theoretically, you can have an economy in which a mining corporation produces and sells iron to a robotics corporation, the robotics corporation produces and sells robots to the mining corporation, which mines more iron, which is used to produce more robots, and so on. These corporations can grow and expand to the far reaches of the galaxy, and all they need are robots and computers – they don't need humans even to buy their products.

Indeed, already today computers and algorithms are beginning to function as clients in addition to producers. In the stock exchange, for example, algorithms are becoming the most important buyers of bonds, shares and commodities. Similarly in the advertisement business, the most important customer of all is an algorithm: the Google search algorithm. When people design Web pages, they often cater to the taste of the Google search algorithm rather than to the taste of any human being.

Algorithms obviously have no consciousness, so unlike human consumers, they cannot enjoy what they buy, and their decisions are not shaped by sensations and emotions. The Google search algorithm cannot taste ice cream. However, algorithms select things based on their internal calculations and built-in preferences, and these preferences increasingly shape our world. The Google search algorithm has a very sophisticated taste when it comes to ranking the Web pages of ice-cream vendors, and the most successful ice-cream vendors in the world are those that the Google algorithm ranks first – not those that produce the tastiest ice cream.

I know this from personal experience. When I publish a book, the publishers ask me to write a short description that they use for publicity online. But they have a special expert, who adapts what I write to the taste of